Pigovian Tax

What Is a Pigovian Tax?

A Pigovian tax is a tax assessed against businesses and private individuals that engage in activities that create adverse side effects for society. Those can include environmental pollution, strains on public healthcare systems from the sale of tobacco products, or any other side effects that have an external, negative impact.

Pigovian taxes were named after English economist Arthur Pigou, a significant contributor to early externality theory.

• In the environmental context: • *"A Pigouvian tax is an emission fee exactly equal to the aggregate marginal damage caused by the emissions at the efficient level of pollution."*

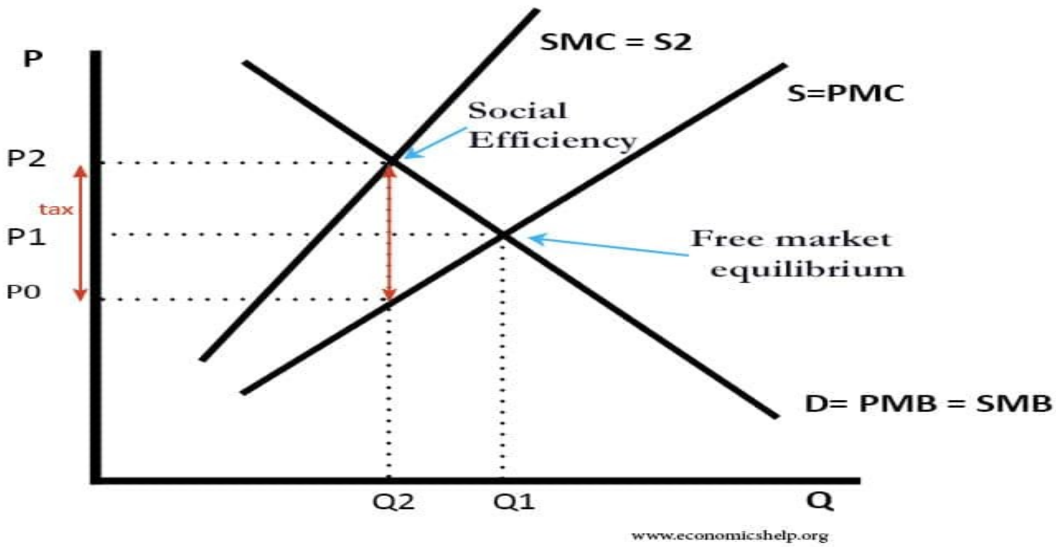
Explaination

In economics, **a negative externality** is a byproduct produced by some individual, business, or industry that has a negative impact on society, but where the entity that created this byproduct does not pay for it. Instead, society pays the price. Examples include **air and noise pollution, toxic runoff, and the inadvertent killing of pollinators through pesticides**, among others.

A Pigovian tax is meant to discourage activities that impose a cost of production onto third parties and society as a whole. According to Pigou, these negative externalities prevent a market economy from reaching equilibrium when producers do not take on all costs of production. This adverse effect might be corrected, he suggested, by levying taxes equal to the externalized costs. Ideally, the tax would also give the producer an incentive to reduce the negative externalities they are responsible for.

Once Pigou factored in external costs to society, the economy suffered **deadweight loss** from excess pollution beyond the "socially optimal" level. Pigou believed that state intervention should correct negative externalities, which he considered **a market failure. He suggested that this be accomplished through taxation.**

Diagram



- In a free market, the equilibrium will be at Q1 – where D=S. At this output, there is social inefficiency.
- At Q1, the social marginal cost (SMC) is greater than the social marginal benefit (SMB) – there is overconsumption.
- If the government place a tax equal to the external marginal cost, then consumers will be paying the full social marginal cost. (SMC)
- This will reduce the demand from Q1 to Q2 and this will be socially efficient because at Q2 (SMC=SMB)

Examples of a Pigovian Tax

Despite any counterarguments toward Pigou's theories, Pigovian taxes are prevalent in society today. One of the most widespread Pigovian taxes is the **carbon emissions tax.** Some governments impose such taxes on any company that burns fossil fuels. When burned, fossil fuels emit greenhouse gases, which cause global warming, damaging the planet in a multitude of ways.

The carbon tax is intended to factor in the real cost of burning fossil fuels, which is paid by society. The end role of the carbon tax is to ensure that the producers of the emissions are the ones incurring this external cost.

Another Pigovian tax, common in Europe, is a **tax on plastic bags**, and sometimes even paper bags. This encourages consumers to bring their own reusable bags from home to discourage the use of plastic and paper. Plastic is a byproduct of burning fossil fuels and results in damage to marine life, while paper bags can exacerbate deforestation. By charging even a small amount, like a few cents per bag, the tax discourages their production.

Taxes on "sin" items like alcohol and cigarettes can also be construed as Pigovian taxes. This is because they discourage behavior that can not only harm the individual user but also have damaging effects on others. Second-hand smoke is an obvious example, but so too is the financial burden on the healthcare system from smokers who become ill with cancer or emphysema. Alcohol is responsible for drunk driving accidents, including injuries and deaths among innocent others.